

PAPER_942: Collimation-Power Mapping

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 213 **Source:** linewidth_jet_modulation.py (CollimationPowerMapping) **Calculator:** CollimationPowerMappingCalc (CP4 #526) **CVW:** v2.0.0 compliant

Abstract

We derive the mapping between phonon linewidth Γ , jet collimation half-angle θ_{half} , and brightness contrast for astrophysical jets. Narrow linewidths ($\Gamma \leq 0.07$ THz) produce tightly collimated jets with $\theta_{\text{half}} \lesssim 3^\circ$, while broad linewidths ($\Gamma > 0.15$ THz) yield wide opening angles $\theta_{\text{half}} \gtrsim 10^\circ$. The contrast (peak-to-background ratio) scales with M_{jet} .

1. Collimation Relation

$$\theta_{\text{half}} = \max\left(0.5^\circ, \frac{30^\circ}{Q}\right)$$

where $Q = \omega_{\text{SCM}}/(2\Gamma)$ is the quality factor.

2. Mapping Results

Γ (THz)	Q	θ_{half}	Contrast
0.05	12.5	2.4°	High
0.10	6.25	4.8°	Moderate
0.30	2.08	14.4°	Low

3. Observational Implications

The Q - θ_{half} relation provides a direct VLBI prediction: systems with narrower phonon linewidths should exhibit tighter jet collimation at mas scales, testable with EHT and ngEHT observations.

4. Source Data

- **File:** linewidth_jet_modulation.py
 - **Session:** 213
 - **CP4 Class:** CollimationPowerMappingCalc (#526)
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References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. Event Horizon Telescope Collaboration (2019) – ApJL, 875, L1